

Installing packages through automated shell script

```
#!/bin/bash

packages=("nginx" "curl" "wget")

apt update

for pkg in "${packages[@]}"
do
    if dpkg -s "$pkg" >/dev/null 2>&1
    then
        echo "$pkg is already installed. Skipping."
    else
        echo "$pkg is not installed. Installing..."
        apt install -y "$pkg"
    fi
done
```

Explanation of each line -

- `packages=("nginx" "curl" "wget")`: Creates an array named `packages` containing three specific software names.
- `apt update`: Updates the local package index to ensure the latest versions are installed.
- `for pkg in "${packages[@]}"`: Starts a loop to iterate through each item in the `packages` array.
- `if dpkg -s "$pkg" >/dev/null 2>&1`: Checks if the package is installed while hiding any error messages.
- `apt install -y "$pkg"`: Installs the missing package automatically without prompting for manual confirmation.

```
root@TCS1INMLJZJJ3: ~  
Preparing to unpack .../5-libnginx-mod-mail_1.18.0-6ubuntu14.12_amd64.deb ...  
Unpacking libnginx-mod-mail (1.18.0-6ubuntu14.12) ...  
Selecting previously unselected package libnginx-mod-stream.  
Preparing to unpack .../6-libnginx-mod-stream_1.18.0-6ubuntu14.12_amd64.deb ...  
Unpacking libnginx-mod-stream (1.18.0-6ubuntu14.12) ...  
Selecting previously unselected package libnginx-mod-stream-geoip2.  
Preparing to unpack .../7-libnginx-mod-stream-geoip2_1.18.0-6ubuntu14.12_amd64.deb ...  
Unpacking libnginx-mod-stream-geoip2 (1.18.0-6ubuntu14.12) ...  
Selecting previously unselected package nginx-core.  
Preparing to unpack .../8-nginx-core_1.18.0-6ubuntu14.12_amd64.deb ...  
Unpacking nginx-core (1.18.0-6ubuntu14.12) ...  
Selecting previously unselected package nginx.  
Preparing to unpack .../9-nginx_1.18.0-6ubuntu14.12_amd64.deb ...  
Unpacking nginx (1.18.0-6ubuntu14.12) ...  
Setting up nginx-common (1.18.0-6ubuntu14.12) ...  
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /lib/systemd/system/nginx.service.  
Setting up libxslt1.1:amd64 (1.1.34-4ubuntu0.22.04.5) ...  
Setting up libnginx-mod-http-geoip2 (1.18.0-6ubuntu14.12) ...  
Setting up libnginx-mod-mail (1.18.0-6ubuntu14.12) ...  
Setting up libnginx-mod-http-image-filter (1.18.0-6ubuntu14.12) ...  
Setting up libnginx-mod-stream (1.18.0-6ubuntu14.12) ...  
Setting up libnginx-mod-http-xslt-filter (1.18.0-6ubuntu14.12) ...  
Setting up libnginx-mod-stream-geoip2 (1.18.0-6ubuntu14.12) ...  
Setting up nginx-core (1.18.0-6ubuntu14.12) ...  
* Upgrading binary nginx  
Setting up nginx (1.18.0-6ubuntu14.12) ...  
Processing triggers for ufw (0.36.1-4ubuntu0.1) ...  
Processing triggers for man-db (2.10.2-1) ...  
Processing triggers for libc-bin (2.35-0ubuntu3.13) ...  
curl is already installed. Skipping.  
wget is already installed. Skipping.  
root@TCS1INMLJZJJ3:~#
```

[OK]

Error Handling

```
root@TCS1INMLJZJJJS3: ~  
#!/bin/bash  
  
set -e  
  
mkdir /tmp/devops-test || echo "Failed to create directory"  
  
cd /tmp/devops-test || echo "Failed to enter directory"  
  
touch test.txt || echo "Failed to create file"  
  
echo "Script completed successfully"  
~  
~  
~
```

Explanation of each line -

- `set -e`: Halts script execution instantly if any command fails with an error.
- `mkdir /tmp/devops-test || echo "Failed to create directory"`: Tries making a folder, printing an error message if it fails
- `cd /tmp/devops-test || echo "Failed to enter directory"`: Tries entering the folder, printing an error message if it fails.
- `touch test.txt || echo "Failed to create file"`: Tries creating an empty file, printing an error if it fails.
- `echo "Script completed successfully"`: Prints a success message to the screen when all steps work.

```
root@TCS1INMLJZJJJS3: ~  
root@TCS1INMLJZJJJS3:~# ./error_handling.sh  
Script completed successfully  
root@TCS1INMLJZJJJS3:~# |
```